From EEG Intent to Simulated Robotic Pickup: A Midterm Project Reflection

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Abstract: This project investigates whether a low-bandwidth motor-imagery electroencephalography (EEG) interface can reliably enable and pause a learned simulated robotic pickup skill. Our initial proposal targeted the evaluation gap between offline neural classification and physical task execution: an above-chance window accuracy does not establish that noisy, flickering predictions can drive contact-rich manipulation within a fixed time budget without unintended motion during rest. We decomposed the system into per-person rest-versus-hand-imagery decoding, causal streaming replay with command gating, and an autonomous manipulation policy wrapped in a project-built `HOLD` and completion-latch adapter. Across a corrected five-person scripted Franka benchmark (15 episodes) and 36 rollouts coupling recorded EEG to an attributed pretrained behavior-cloning recurrent policy (`BC-RNN`), the pipeline achieved 21/27 retained lifts on three held-out participants (`S001`–`S003`) and 3/9 on an exploratory participant (`S006`). Crucially, physical outcomes separated into two failure regimes governed by an empirical minimum activation threshold of $(59, 120]$ in-window commands: strict gates starved recall-limited decoders (`S002`: 3/3 raw vs. 0/3 filtered), whereas precision-limited decoders completed 9/9 rollouts by false-triggering across 93.3% of rest windows (`S003`). For the remainder of the semester, we drop custom from-scratch reinforcement learning to execute a frozen, matched-control study on uninspected participants (`S011`–`S015`) pairing task completion with rest-motion and calibration-bounded gate evaluation.

1 Introduction

Non-invasive EEG offers an accessible assistive command channel, yet motor-imagery signals exhibit low signal-to-noise ratios, nonstationarity, and pronounced cross-subject variability [1, 2]. Standard BCI benchmarks demonstrate that offline accuracy frequently fails to transfer across splitting choices or cohorts [3], and an isolated classifier score leaves a critical robotics question unanswered: *when predictions stream causally into a stateful controller, do stabilization filters improve task completion, or do they starve the robot of the sustained activation required to grasp?* Our original twelve-week proposal therefore aimed to build a reproducible, simulation-first evaluation harness connecting public EEG to simulated robotic manipulation rather than claiming a live or clinical device.

Original gap, insight, and plan. The proposal identified that training an end-to-end policy directly from noisy EEG would compound neural nonstationarity with contact discovery. Our core insight was hierarchical shared autonomy [4]: restrict the decoder to a conservative binary choice—instructed rest (`T0`) versus combined left/right fist imagery (`T1`/`T2`) in the PhysioNet EEG Motor Movement/Imagery dataset [5]—pass streaming probability estimates through a command gate (`ENABLE` vs. `HOLD`), and delegate continuous joint control to a low-level policy. Mapping fist imagery to a pickup request is an explicit interface convention, not evidence that the dataset encodes native grasp intent. Our twelve-week schedule planned leakage-safe Common Spatial Patterns (CSP) [6], Riemannian [7], and EEGNet baselines (Weeks 1–6), causal replay and temporal command filtering (Weeks 7–8),

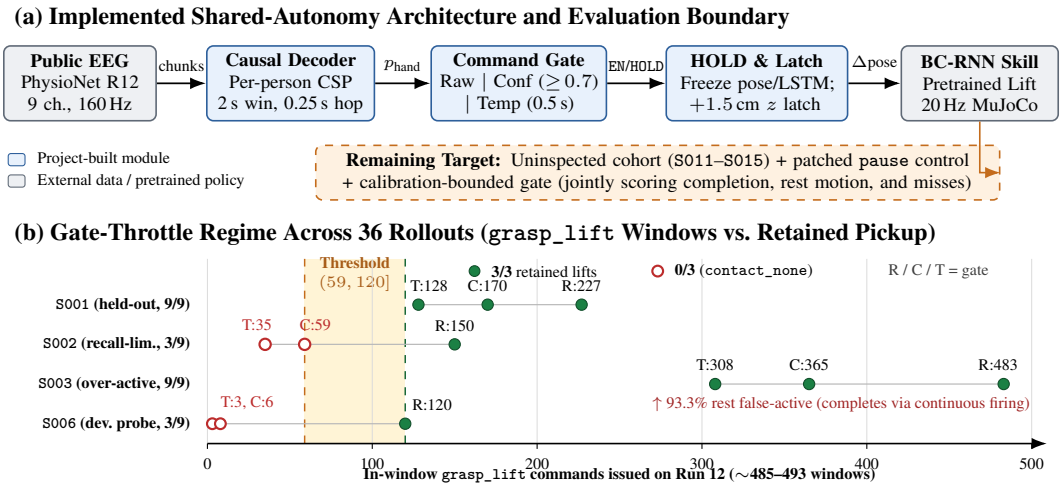


Figure 1: **System overview and midterm empirical regime split. (a)** Project-built modules (blue) connect public motor-imagery EEG to an externally pretrained `robomimic` `BC-RNN` policy (gray) via causal decoding, command gating, and a `HOLD`-preserving completion latch; orange indicates the remaining prospective benchmark. **(b)** Across 36 learned-policy rollouts (3 scene seeds per gate), cube acquisition and terminal retention separate cleanly across an empirical command-count threshold of $(59, 120]$ `grasp_lift` windows, exposing a tradeoff between strict-gate starvation (`S002`, `S006`) and unbounded rest false activation (`S003`).

Isaac Lab reinforcement learning (RL) pickup training (Weeks 9–10), and matched end-to-end trials (Weeks 11–12). Figure 1 illustrates the implemented architecture alongside our central empirical finding at midterm.

2 Reflection: Where are we?

EEG decoding and scripted baseline. We built a leakage-safe MNE preprocessing and causal replay pipeline calibrating per-person models on runs 4 and 8 with run 12 held out (9 sensorimotor channels, 8–30 Hz, 2.0 s windows). On participants `S001`–`S005` (150 test windows), individual CSP/LDA reached 70.67% balanced accuracy (29.33% rest false activation), whereas subject-pooled CSP/LDA dropped to 49.33%; Riemannian tangent-space classification reached 72.67% individual and 70.00% pooled accuracy. On a separate five-person cohort (`S006`–`S010`), individual CSP/LDA scored 59.33% balanced accuracy (29.33% rest false activation). Driving a frozen *scripted* Franka lift over 120-second replays after fixing an initial `HOLD`-drift bug, raw, confidence (≥ 0.7), and temporal (0.5 s persistence) gates achieved physical lifts in 5/5, 3/5, and 2/5 subjects, respectively. Filtering reduced mean rest command activation from 38.7% to 25.7% and 19.6% (and physical rest motion from 13.5% to 9.1%), but increased missed imagery bouts from 12 to 30 and 38.

Robot learning progression and pivot. Custom policy learning in Isaac Lab proved substantially harder than anticipated. While our PPO agent lifted from privileged in-hand resets, ready-pose table transfer yielded 0/5 pickups. Subsequent curriculum variants reached 2/5 lifts in a fixed near-table progress screen (with one drop and asymmetric right-finger contact peaks of 92–169 N that knocked the cube sideways) and peaked at 3/5 sustained bilateral acquisitions in a unilateral near-table screen, still short of reliable ready-pose pickup or validated `HOLD` invariance. Rather than stalling the interface study on fragile reward shaping, we integrated an attributed, externally pretrained `robomimic` `BC-RNN` Panda Lift policy [8] in native MuJoCo. Naive intermittent gating initially retained the cube in 0/5 terminal states; engineering a stateful adapter that freezes both the end-effector target and recurrent hidden state during `HOLD`, paired with a one-time +1.5 cm world-z goal offset at bilateral grasp acquisition, achieved 21/21 retained lifts across positive development controls and 0/3 on `always-hold` negatives.

Connected EEG-to-learned-policy results. We then executed 36 offline decision-stream rollouts (400 steps / 20 s at 20 Hz; seeds 0–2 per gate) across exploratory subject S006 and all three held-out subjects S001–S003, bracketed on S001 by matched `always-enable` (3/3 retained, steps 47–58) and `always-hold` (0/3) controls:

- **Balanced held-out (S001, 76.7% BA, 20.0% rest false-active):** Completed and retained **9/9** lifts across raw (227 commands), confidence (170), and temporal (128) gates with zero drops.
- **Recall-limited (S002, 73.3% BA, 53.3% hand recall; and exploratory S006):** Completed **3/9** each—retaining 3/3 under raw commands (S002: 150; S006: 120) but 0/3 under confidence (S002: 59; S006: 6) and temporal (S002: 35; S006: 3) gates, which starved the policy of reach time.
- **Precision-limited (S003, 53.3% BA, 93.3% rest false-active):** Completed **9/9** lifts across all three gates (483/365/308 commands), succeeding physically because near-continuous false firing acted like an always-on trigger.

Was the gap real, did we verify our insight, and what was harder than expected? The gap between offline decoding and closed-loop utility proved sharper than hypothesized, though as an empirical systems problem rather than an unstudied paradigm [4, 9]. Uncalibrated LDA posteriors caused fixed-threshold (≥ 0.7) duty cycles to swing nearly $50\times$ across individuals (0.012 on S006 vs. 0.565 on S008), making offline accuracy a poor predictor of completion: S002 (73.3% BA) failed filtered gates, whereas S003 (53.3% BA) completed every rollout via unintended rest activation. Our shared-autonomy insight is verified in bounded form—binary EEG replay can drive a learned manipulation skill when mediated by a recurrent-state-preserving HOLD and grasp latch—while falsifying our naive assumption that fixed temporal filtering uniformly helps. Three engineering subtleties were harder than expected: sustaining bilateral contact through terminal states rather than brief reward spikes, preventing HOLD target drift, and auditing control baselines—where a code review caught and patched an inverted `pause`-matched control branch (`hold` vs. `policy` inside the timed window) before any invalid causal localization was claimed.

3 Updated work plan

To guarantee a rigorous, self-contained project by semester end, we pivot from training manipulation policies from scratch to a controlled shared-autonomy systems benchmark around the attributed BC-RNN controller. Because fixed confidence thresholds either starve conservative decoders or reward over-active ones, we also add a lightweight, calibration-only quantile/conformal gate threshold (fit strictly on runs 4 and 8 to target bounded rest false activation) alongside our frozen gates.

Table 1: Revisions from the initial twelve-week proposal to the final semester plan.

Original component	Status	Revision and technical rationale
EEG baselines & replay	Complete	Per-person/pooled CSP and Riemannian pipelines plus causal replay verified; retain binary T0 vs. T1/T2 labels.
Isaac Lab PPO table pickup	Stopped	Best near-table policy reached 3/5 lifts with asymmetric impact forces; retire custom RL and use attributed `robomimic` BC-RNN.
HOLD & retention adapter	Complete	Added state-freezing HOLD executor and $+1.5$ cm latch (21/21 dev. controls); patched inverted `pause`-control bug.
Learned-policy EEG replay	Held-out (S001–S003)	24/36 retained lifts across S001–S003 and S006 revealed the $(59, 120]$ trigger threshold; expand to uninspected cohort.
Fine gestures & hardware	Deferred	Dataset lacks grasp/wrist labels; replace extra deep architectures with calibration-bounded gating and matched controls.

Remaining schedule (Weeks of Nov. 9 – Dec. 7). (1) Nov. 9: Execute the corrected `pause`-matched timing controls on `S001`–`S003` to isolate duty-cycle sufficiency from fragmented EEG timing, and freeze all code hashes, unseen scene seeds, and per-person calibration rules (runs 4/8 only) before opening new subjects. **(2) Nov. 16–23:** Run a single prospective evaluation on five uninspected PhysioNet participants (`S011`–`S015`, run 12) comparing raw, fixed-confidence, temporal, and calibration-bounded gates against `always-hold`, `always-enable`, and duration-matched `pause` controls without post-hoc window selection. **(3) Nov. 30:** Join EEG cue annotations to MuJoCo telemetry to jointly score retained completion, robot motion during annotated rest, missed imagery bouts, and detection latency, testing whether calibration-bounded gating curbs `S003`-style rest motion without inducing `S002`-style starvation. **(4) Dec. 7:** Deliver the final report, synchronized trace-and-rollout video, and one-command reproduction scripts. If uninspected subjects fail to acquire, we will report the exact failure mechanism without relaxing thresholds; offline replay will not be represented as live closed-loop BCI or clinical validation.

4 AI statement

AI coding and writing assistants (OpenAI Codex, Claude, and Google Antigravity) were used to help organize literature metadata, scaffold simulation/evaluation scripts, audit control harnesses, summarize logged telemetry, and refine the prose and TikZ figure in this report. All numerical claims, SHA-256 artifact records, and experimental interpretations were verified against saved repository outputs, externally pretrained models are explicitly attributed, and full responsibility for the technical conclusions and limitations rests with the author.

References

[1] D. J. McFarland and J. R. Wolpaw. EEG-based brain–computer interfaces. *Current Opinion in Biomedical Engineering*, 4:194–200, 2017. doi:10.1016/j.cobme.2017.11.004.

[2] F. Lotte, L. Bougrain, A. Cichocki, M. Clerc, M. Congedo, A. Rakotomamonjy, and F. Yger. A review of classification algorithms for EEG-based brain–computer interfaces: A 10 year update. *Journal of Neural Engineering*, 15(3):031005, 2018. doi:10.1088/1741-2552/aab2f2.

[3] V. Jayaram and A. Barachant. MOABB: Trustworthy algorithm benchmarking for BCIs. *Journal of Neural Engineering*, 15(6):066011, 2018. doi:10.1088/1741-2552/aadea0.

[4] I. Akinola, B. Wang, J. Varley, and P. K. Allen. Task level hierarchical system for BCI-enabled shared autonomy. In *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, pages 219–225, 2017. doi:10.1109/HUMANOIDS.2017.8246878.

[5] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer, and J. R. Wolpaw. EEG motor movement/imagery dataset (version 1.0.0). PhysioNet, 2009. doi:10.13026/C28G6P.

[6] B. Blankertz, R. Tomioka, S. Lemm, M. Kawanabe, and K.-R. Müller. Optimizing spatial filters for robust EEG single-trial analysis. *IEEE Signal Processing Magazine*, 25(1):41–56, 2008. doi:10.1109/MSP.2008.4408441.

[7] A. Barachant, S. Bonnet, M. Congedo, and C. Jutten. Multiclass brain–computer interface classification by Riemannian geometry. *IEEE Transactions on Biomedical Engineering*, 59(4): 920–928, 2012. doi:10.1109/TBME.2011.2172210.

[8] A. Mandlekar, D. Xu, J. Wong, S. Nasiriany, C. Wang, R. Kulkarni, L. Fei-Fei, S. Savarese, Y. Zhu, and R. Martín-Martín. What matters in learning from offline human demonstrations for robot manipulation. In *Conference on Robot Learning (CoRL)*, pages 1678–1690, 2021.

[9] I. Carrara and T. Papadopoulo. Pseudo-online framework for BCI evaluation: A MOABB perspective. *arXiv preprint arXiv:2308.11656*, 2023.